

Assignment No 27

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Domain : AI &ML

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Q1.

```
... First 10 Rows:  
      Feature1  Feature2  
0  0.830586 -0.447733  
1  0.701678  0.816918  
2  1.022080 -0.492571  
3 -0.316765  0.953438  
4  0.293226  1.057185  
5  1.254903 -0.483435  
6  1.053851 -0.442690  
7  0.031380  1.053012  
8  0.620545  0.713096  
9  1.787657 -0.142959
```

Shape of the dataset: (500, 2)

Q2.

```
▶ # Import StandardScaler
from sklearn.preprocessing import StandardScaler
import pandas as pd

# Create StandardScaler object
scaler = StandardScaler()

# Scale the data
X_scaled = scaler.fit_transform(df)

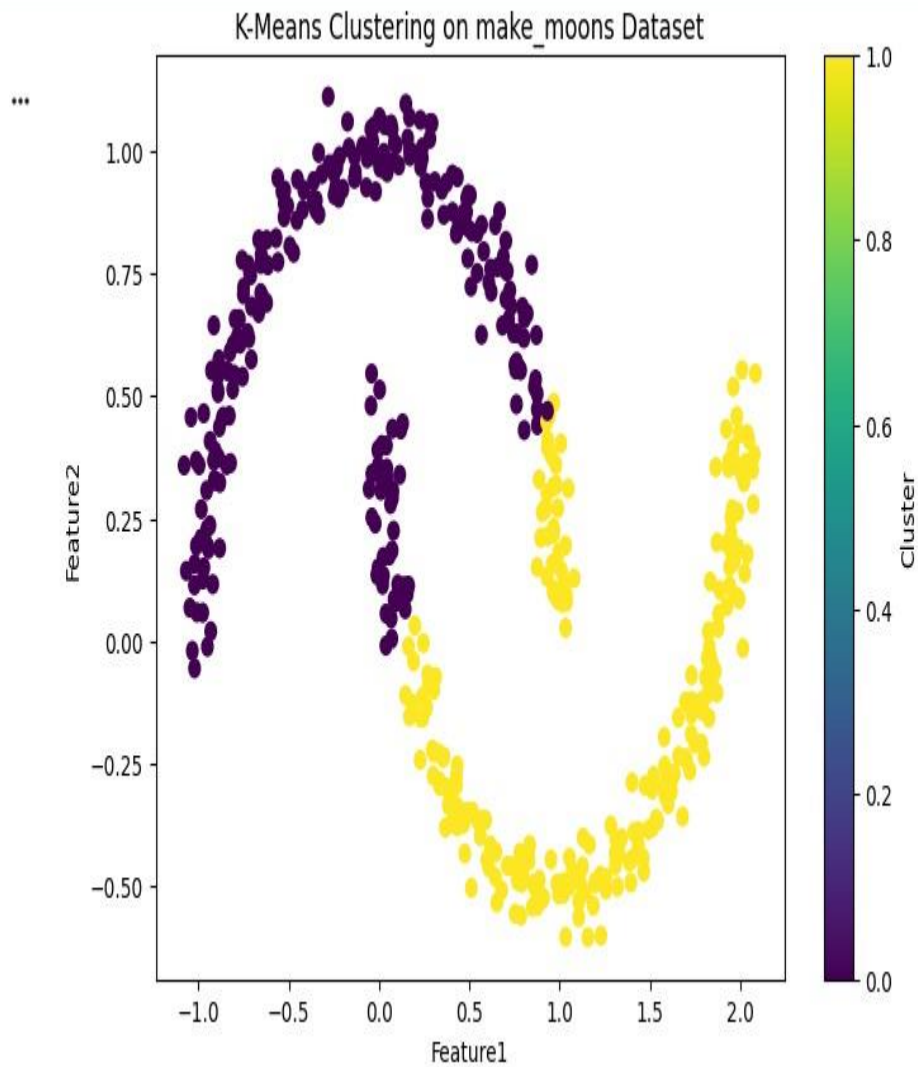
# Display the first 5 rows of the scaled data
scaled_df = pd.DataFrame(X_scaled, columns=["Feature1", "Feature2"])

print("First 5 Rows of Scaled Data:")
print(scaled_df.head())
```

```
... First 5 Rows of Scaled Data:
   Feature1  Feature2
0  0.375125 -1.403679
1  0.226815  1.135628
2  0.595442 -1.493710
3 -0.944918  1.409748
4 -0.243115  1.618062
```

Q3.

```
***      Feature1  Feature2  kmeans_cluster
0  0.830586 -0.447733          1
1  0.701678  0.816918          0
2  1.022080 -0.492571          1
3 -0.316765  0.953438          0
4  0.293226  1.057185          0
```

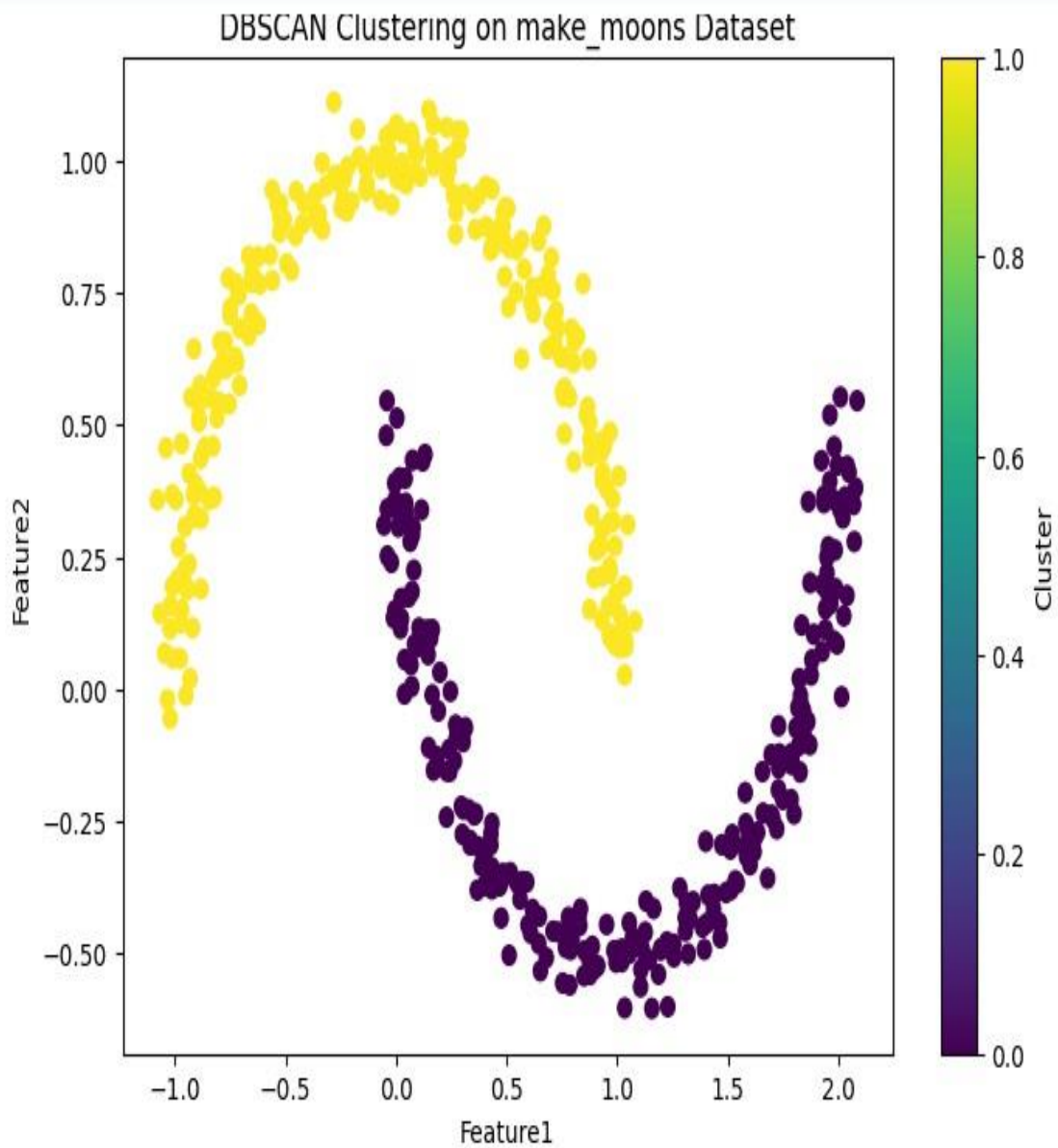


Q4.

```
...   Feature1  Feature2  kmeans_cluster  dbscan_cluster
0    0.830586 -0.447733                1                0
1    0.701678  0.816918                0                1
2    1.022080 -0.492571                1                0
3   -0.316765  0.953438                0                1
4    0.293226  1.057185                0                1
5    1.254903 -0.483435                1                0
6    1.053851 -0.442690                1                0
7    0.031380  1.053012                0                1
8    0.620545  0.713096                0                1
9    1.787657 -0.142959                1                0
```

Unique Cluster Labels: [0 1]

...



Q5.

Observations

KMeans

KMeans divides the dataset into circular (spherical) clusters.

On the make_moons dataset, it fails to separate the two moon-shaped clusters correctly.

Some points from one moon are assigned to the other cluster because KMeans uses the distance from cluster centroids.

It cannot identify noise or outliers.

DBSCAN

DBSCAN groups points based on density instead of distance from a centroid.

It successfully detects the two moon-shaped clusters.

It does not require specifying the number of clusters beforehand.

If there are any outliers, DBSCAN labels them as -1 (noise) instead of forcing them into a cluster.

Q6.

```
from sklearn.cluster import DBSCAN
import pandas as pd

eps_values = [0.2, 0.3, 0.4, 0.5]

results = []

for eps in eps_values:
    dbscan = DBSCAN(eps=eps, min_samples=5)
    labels = dbscan.fit_predict(X_scaled)

    n_clusters = len(set(labels)) - (1 if -1 in labels else 0)

    n_noise = list(labels).count(-1)

    results.append([eps, n_clusters, n_noise])

result_df = pd.DataFrame(
    results,
    columns=["eps", "Number of Clusters", "Number of Noise Points"]
)

print(result_df)
```

```
...      eps  Number of Clusters  Number of Noise Points
0  0.2          2                0
1  0.3          2                0
2  0.4          2                0
3  0.5          2                0
```


Q7.

```
*** First 5 Rows:
```

	Feature1	Feature2	Feature3	Feature4	Feature5	Feature6
0	-7.430902	6.549456	2.166481	3.468039	-10.240054	8.934439
1	-0.746455	-5.467049	4.342113	-6.163268	-4.821673	-3.392003
2	-2.337856	9.239739	4.094461	1.887751	-6.417925	-9.445362
3	-7.659628	7.961709	0.571836	4.089885	-9.593714	7.660650
4	-10.169566	6.667335	3.105872	3.602202	-6.997864	8.798743

```
Dataset Shape: (500, 6)
```

```
First 5 Rows of Scaled Data:
```

	Feature1	Feature2	Feature3	Feature4	Feature5	Feature6
0	-1.052513	0.719083	0.350444	0.989657	-1.619597	1.414499
1	0.140795	-1.040620	0.841574	-0.857108	0.458731	-0.556293
2	-0.143302	1.113049	0.785668	0.686643	-0.153543	-1.524124
3	-1.093345	0.925894	-0.009533	1.108893	-1.371680	1.210841
4	-1.541419	0.736345	0.562503	1.015382	-0.375990	1.392803

Q8.

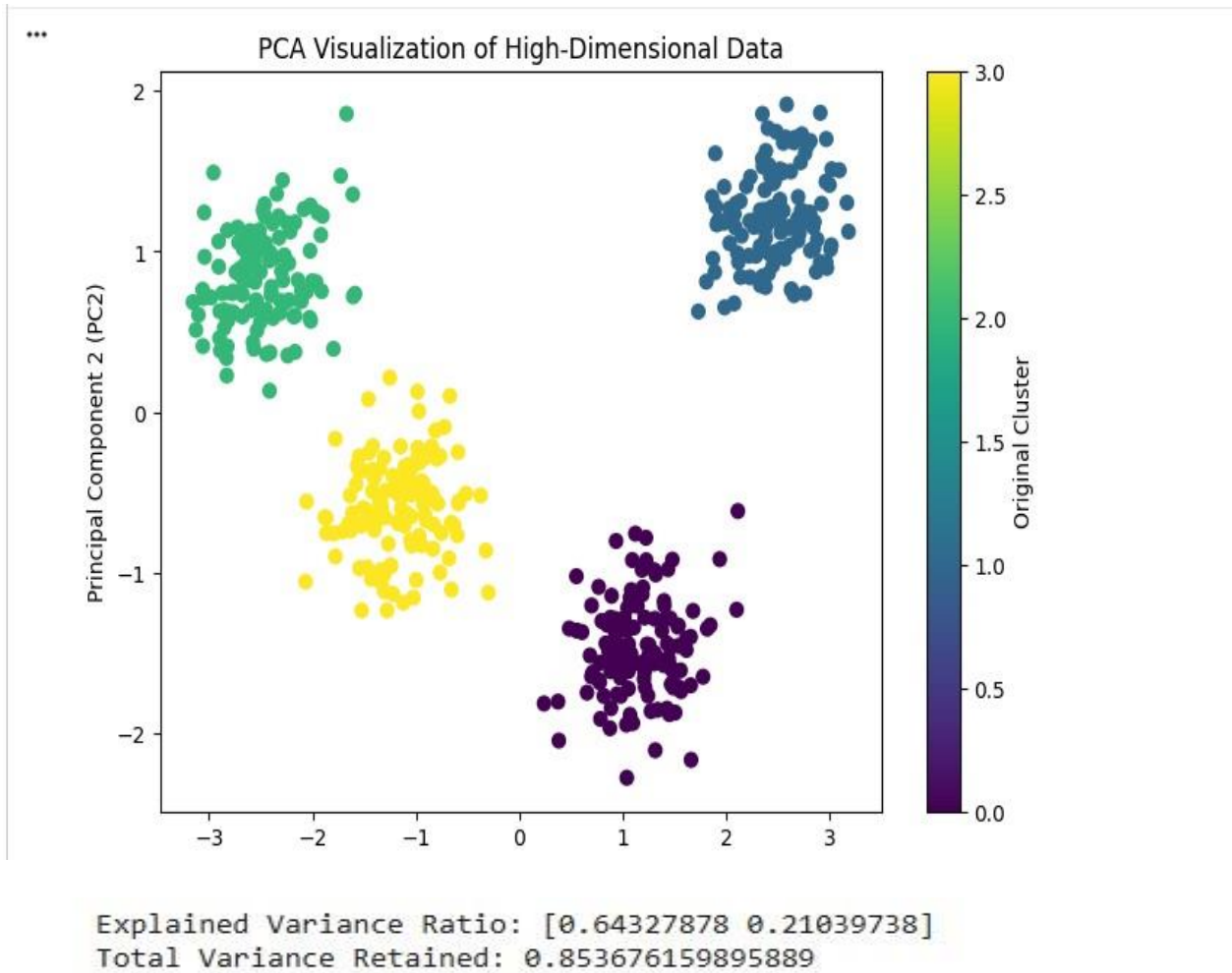
*** First 5 Rows:

	PC1	PC2	Cluster
0	2.369765	1.259783	1
1	-0.943367	-0.829777	3
2	0.982191	-1.768556	0
3	2.265086	1.190377	1
4	2.142129	0.840804	1

Explained Variance Ratio:

[0.64327878 0.21039738]

Q9.



Q10.

```
*** First 5 Rows:
      sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0          5.1           3.5           1.4           0.2
1          4.9           3.0           1.4           0.2
2          4.7           3.2           1.3           0.2
3          4.6           3.1           1.5           0.2
4          5.0           3.6           1.4           0.2

Cluster Labels:
DBSCAN_Cluster
1    97
0    49
-1    4
Name: count, dtype: int64

Explained Variance Ratio:
[0.72962445 0.22850762]
```

...

